PAPER_162 — Solar Cycle UQFF: omega_c = 2π/(11yr), Time-Varying B(t), delta_def Defect Factor

Session: 47 | **Date:** March 13, 2026 | **Thread:** 7f9068 | **Domain:** §2.3

Abstract

This paper establishes **per-body cycle frequency ω_c** as a new first-class UQFF parameter, replacing the static magnetic field B_s with a time-varying field B(t) in the Ug1 magnetic dipole term. The 11-year solar cycle drives B(t) = B_s + 0.4·sin(ω_c·t) for the Sun, while each planet uses its own orbital/rotation period. A new Ug1 **defect factor** δ_def = 0.01 modulates the magnetic dipole oscillation amplitude. This paper is the theoretical foundation for PAPER_157's per-body ω_c parameters.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0×10^{-}4 day^{-1}, [SSq] = 0.57) uniquely enabling this analysis establishing a new connection in the UQFF framework not present in Standard Model treatments.

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1. Motivation

Previous UQFF implementations used static magnetic field B_s. The Sun's magnetic field varies by $\pm$40% over the 11-year solar cycle (B range: ~0.6 G to ~2 G at solar average). This time variation affects: - Ug1 (magnetic dipole term) via μ_s(t) = B(t)·Rs^3 - Ug3 (string rotation) via ω'_s(t) = ω_s + ω_c·cos(ω_c·t) - Solar wind velocity $\rightarrow$ δ_sw term in Ug2

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2. Time-Varying Magnetic Field

$$\boxed{B(t) = B_s + 0.4 \cdot \sin(\omega_c \cdot t) + S_{Cm,contrib}}$$

where $S_{Cm,contrib}$ is the superconductive medium contribution (perturbative, typically ~B_s/100).

2.1 Magnetic Dipole Moment μ_s(t)

$$\mu_s(t) = B(t) \cdot R_s^3$$

Body	B_s [T]	ΔB/B_s	Period	ω_c [rad/s]
Sun	1×10^{-4}	40%	11 yr	2π/(11·365.25·86400)
Earth	3×10^{-5}	varies	1 yr (proxy)	2π/(1·365.25·86400)
Jupiter	4×10^{-4}	varies	11.86 yr	2π/(11.86·365.25·86400)
Neptune	1×10^{-4}	varies	164.8 yr	2π/(164.8·365.25·86400)

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3. Modified Ug1 with Defect Factor

$$\boxed{U_{g1}(r,t) = k_1 \cdot \mu_s(t) \cdot \nabla \frac{M_s}{r} \cdot e^{-\alpha t} \cos(\pi t_n) \cdot (1 + \delta_{def} \cdot \sin(0.001t))}$$

where the **defect factor** δ_def = 0.01 introduces a slow perturbation at 0.001 rad/s (~6.3 second period) representing surface magnetic flux tube defects.

Physical basis for δ_def: Magnetic flux tube emergence/submergence at the solar surface creates ~1% modulation in the effective dipole moment on timescales of seconds to minutes (observed in solar magnetogram data).

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4. Modified Ug3 with Time-Varying Rotation

$$\omega_s'(t) = \omega_s + \omega_c \cdot \cos(\omega_c \cdot t)$$

The rotation frequency itself becomes time-modulated by the solar cycle, affecting:

$$U_{g3}(r,t) = k_3 \cdot B_j(t) \cdot \cos(\omega_s'(t) \cdot t \cdot \pi) \cdot P_{core} \cdot E_{react}$$

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5. Per-Body ω_c Implementation (C++)

```cpp
// omega_c assignments (radians per second)
const double YEAR = 365.25 * 86400.0;

body_sun.omega_c     = 2.0 * M_PI / (11.0   * YEAR);  // 11-year solar cycle
body_earth.omega_c   = 2.0 * M_PI / (1.0    * YEAR);  // annual orbital
body_jupiter.omega_c = 2.0 * M_PI / (11.86  * YEAR);  // 11.86-year orbital
body_neptune.omega_c = 2.0 * M_PI / (164.8  * YEAR);  // 164.8-year orbital

// Compute time-varying B
double compute_B_t(const CelestialBody& body, double t) {
    double SCm_contrib = body.SCm_density * 1e-10;  // perturbative small
    return body.Bs_avg + 0.4 * std::sin(body.omega_c * t) + SCm_contrib;
}
```

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6. Testable Predictions

At solar maximum (B_s + 0.4 = 1.4×10^{-4} T vs minimum B_s - 0.4 = 0.6×10^{-4} T):

$$\frac{U_{g1}^{max}}{U_{g1}^{min}} = \frac{1.4}{0.6} \approx 2.33$$

Solar UQFF field strength varies by factor ~2.3 over the solar cycle. This 11-year modulation should correlate with: - Cosmic ray flux variation (observed: ~10-20%) - Interplanetary field modulation (observed) - Long-term climate variation (Maunder Minimum connection)

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7. CP Integration

CP2 update: `UQFFSolarCycleCalculator` — add `omega_c` parameter, `compute_B_t()`, `delta_def` parameter, modified Ug1 and Ug3 equations with time-varying components.

Status: Complete | **CP Stage:** CP2 **Supersedes:** N/A (extends static B_s) | **Related:** PAPER_157 (per-body ω_c usage), PAPER_027 (5-freq resonance including solar), PAPER_086 (Ug1 derivation)

UQFF computed: Solar wind UQFF correction = [SSq]exp(-?r/v) = 5.7e-1exp(-5.0e-4(1AU/400km/s)) = 5.7e-1exp(-3.2e-3) 5.7e-1; dominant at r < 1AU.